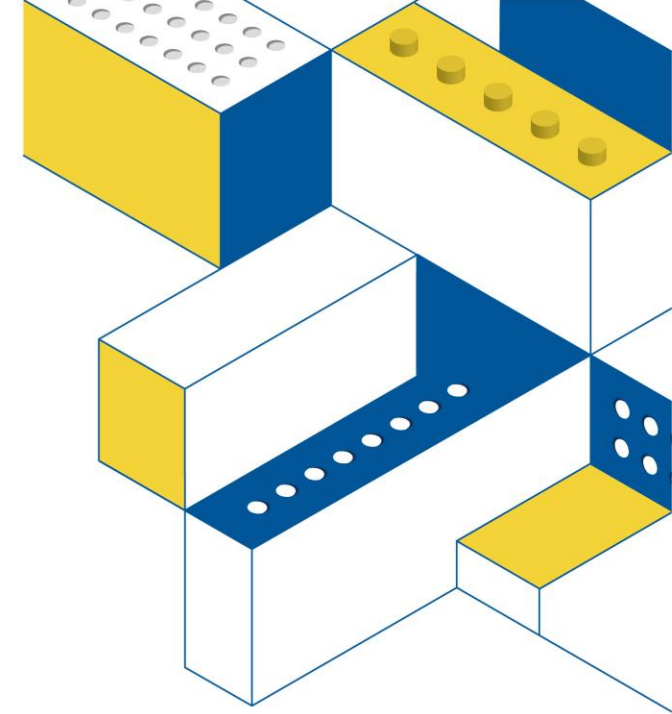


# MATRIXblock Software introduction

# Learning objectives

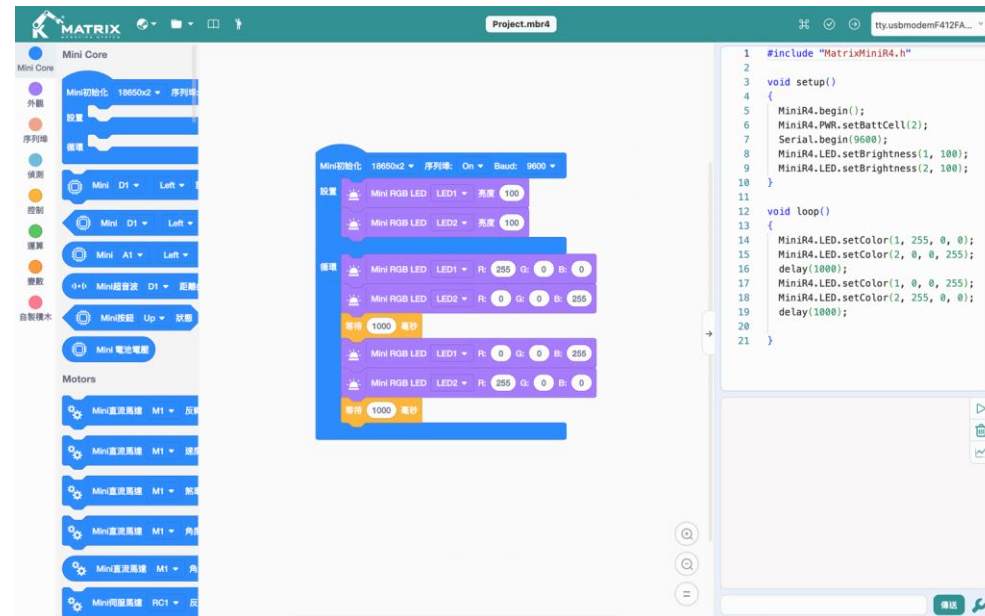
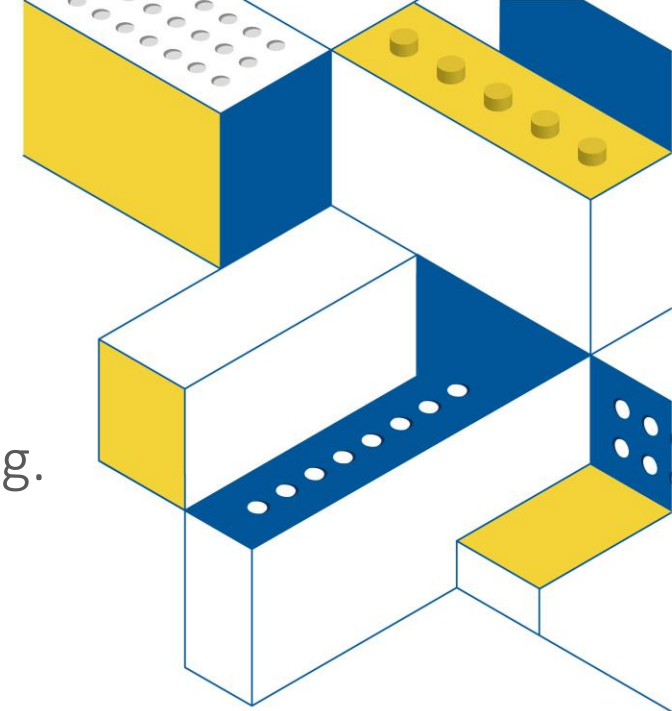
- Getting to know MATRIXblock
- How to connect devices
- How to upload programs



# What is MATRIXblock

MATRIXblock is a robot programming tool based on Scratch. It offers block to C++ previews and serial port monitor for easy data debugging.

This software serves as a perfect bridge from block-based to text-based coding, making it ideal for beginners and educators. With MATRIXblock, users can seamlessly transition to advanced coding while bringing their creative ideas to life.



# Interface Guide



1. Language: Tap to change the language.
2. Files: Add, open, save files, and firmware update.
3. Teaching: Basic introduction, assembly skills, sample models, sample programs.
4. File name: The name of the current file.
5. Compile verification: Compile and validate programs without equipment.
6. Compile and download the program to MATRIX Mini.
7. Robot connection port.

# Interface Guide-Blocks Area

Find blocks by category and color.

The screenshot displays the Scratch Blocks area for a Mini Core project. On the left, a vertical sidebar lists categories with colored circular icons: Mini Core (blue), Looks (purple), Serial (brown), Sensing (teal), Control (orange), Operators (green), Variables (light orange), and My Blocks (pink). The main area shows a grid of blocks. The 'Mini Core' category is highlighted with a yellow border. It contains several blocks: 'Mini Begin' (18650x2 UART), 'Setup' (with a dropdown menu), 'Loop' (with a dropdown menu), 'Mini D1 Left' (with a dropdown menu), 'Mini D1 Left' (with a dropdown menu), 'Mini A1 Left' (with a dropdown menu), 'Mini Ultrasonic D1' (with a dropdown menu), 'Mini Button Up' (with a dropdown menu), 'Mini Battery Voltage', and a 'Motors' section with 'Mini DC Motor M1' (with a dropdown menu), 'Mini DC Motor M1' (with a dropdown menu), 'Mini DC Motor M1' (with a dropdown menu), 'Mini DC Motor M1' (with a dropdown menu), 'Mini DC Motor M1' (with a dropdown menu), and 'Mini Servo Motor RC1' (with a dropdown menu). On the right, a code editor shows the following code:

```
1 #include "MatrixMiniR4.h"
2
3 void setup()
4 {
5
6 }
7
8 void loop()
9 {
10
11 }
```

At the bottom right, there is a 'Send' button and a wrench icon.

# Interface Guide-Edit area

Drag and drop the blocks to this area for programming.

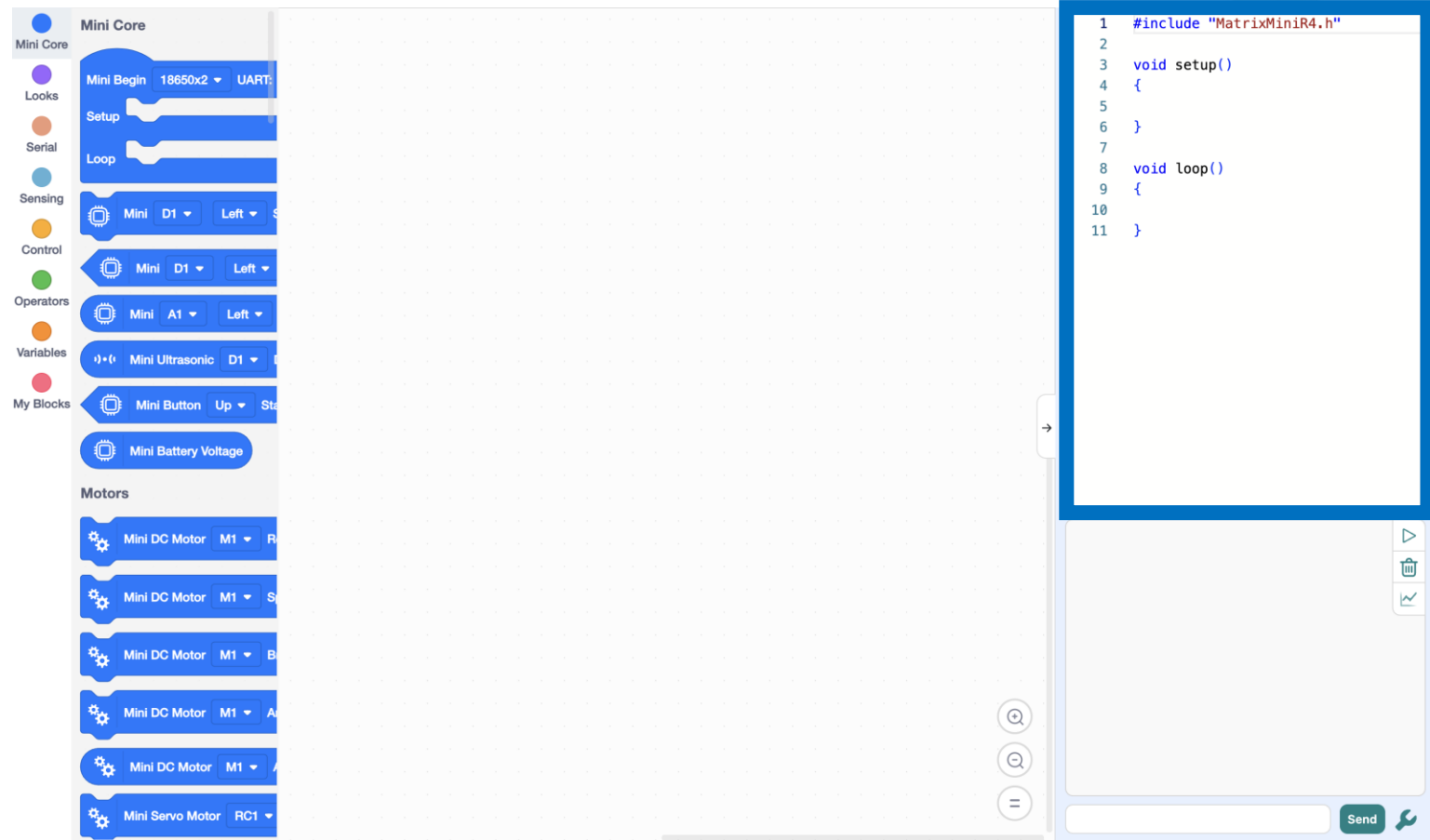
The image shows a programming interface for a Mini Core microcontroller. On the left is a block palette with categories: Mini Core, Looks, Serial, Sensing, Control, Operators, Variables, and My Blocks. The Mini Core category is expanded, showing blocks for Mini Begin, Setup, Loop, Mini D1, Mini A1, Mini Ultrasonic, Mini Button, Mini Battery Voltage, and a Motors section with Mini DC Motor and Mini Servo Motor blocks. The central workspace is a large grid area with a green border, intended for dragging and dropping blocks. On the right is a code editor showing the following C++ code:

```
1 #include "MatrixMiniR4.h"
2
3 void setup()
4 {
5
6 }
7
8 void loop()
9 {
10
11 }
```

At the bottom right of the code editor are icons for running, saving, and undo/redo, along with a 'Send' button and a wrench icon.

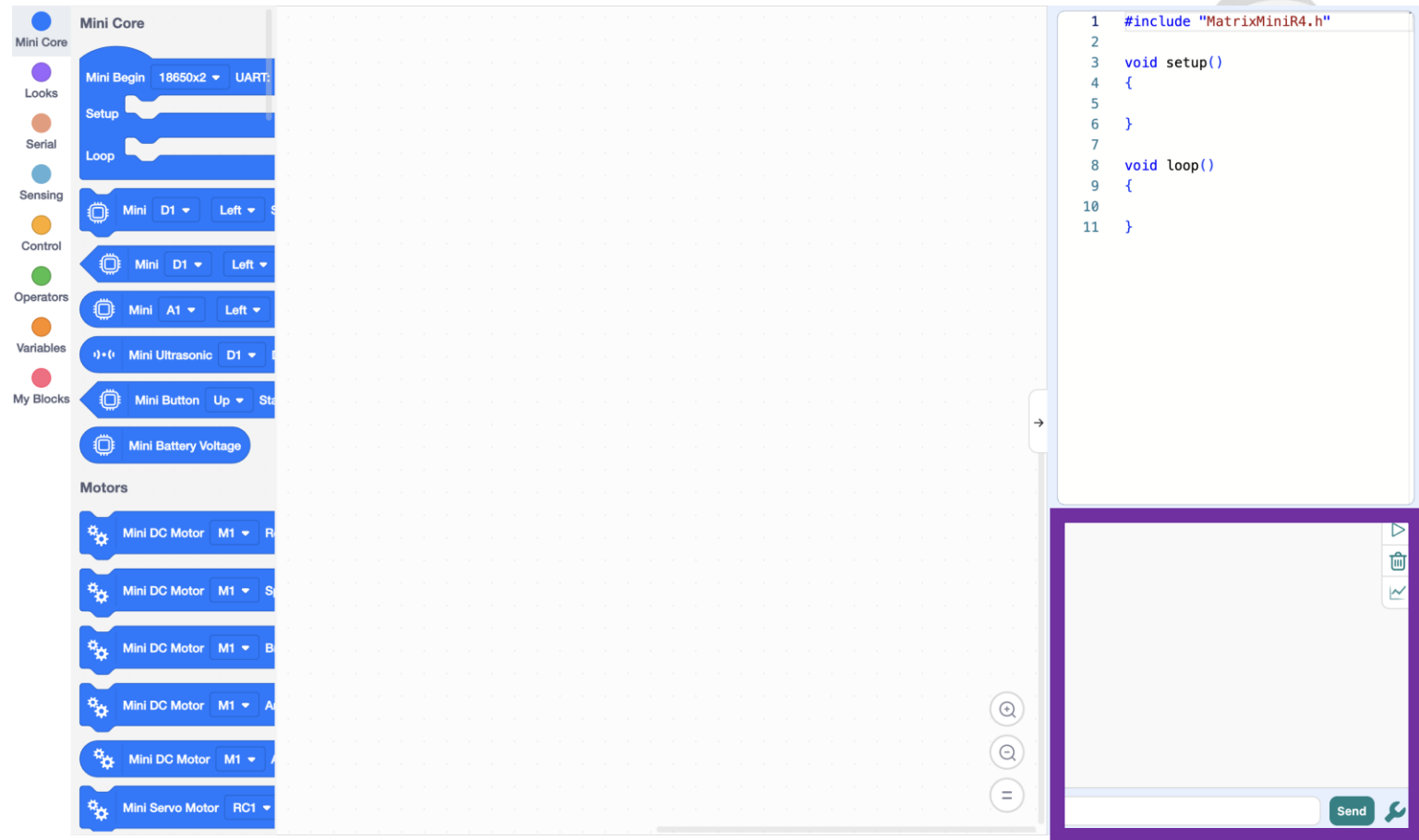
# Interface Guide- Code View

You can see the C++ code that the blocks have been converted to, or copy it to the Arduino IDE for further editing.



# Interface Guide- Information Window

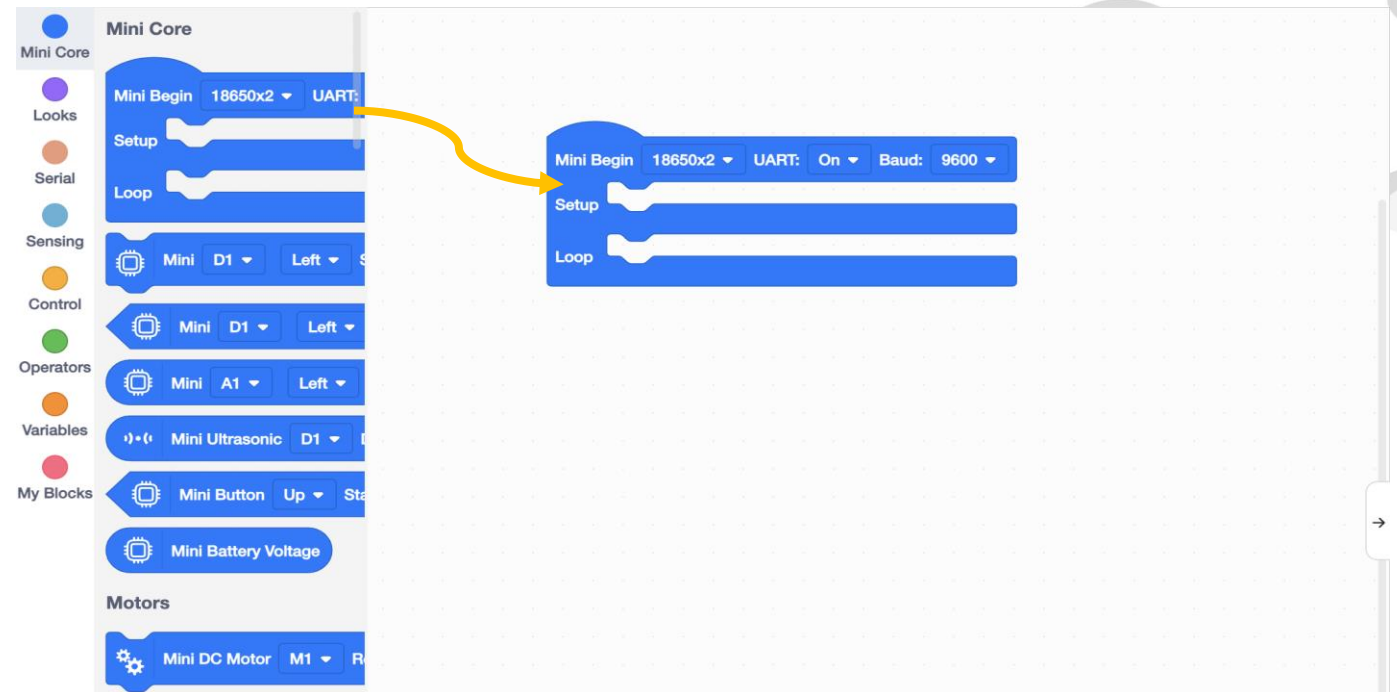
Use the serial port to communicate with the robot.





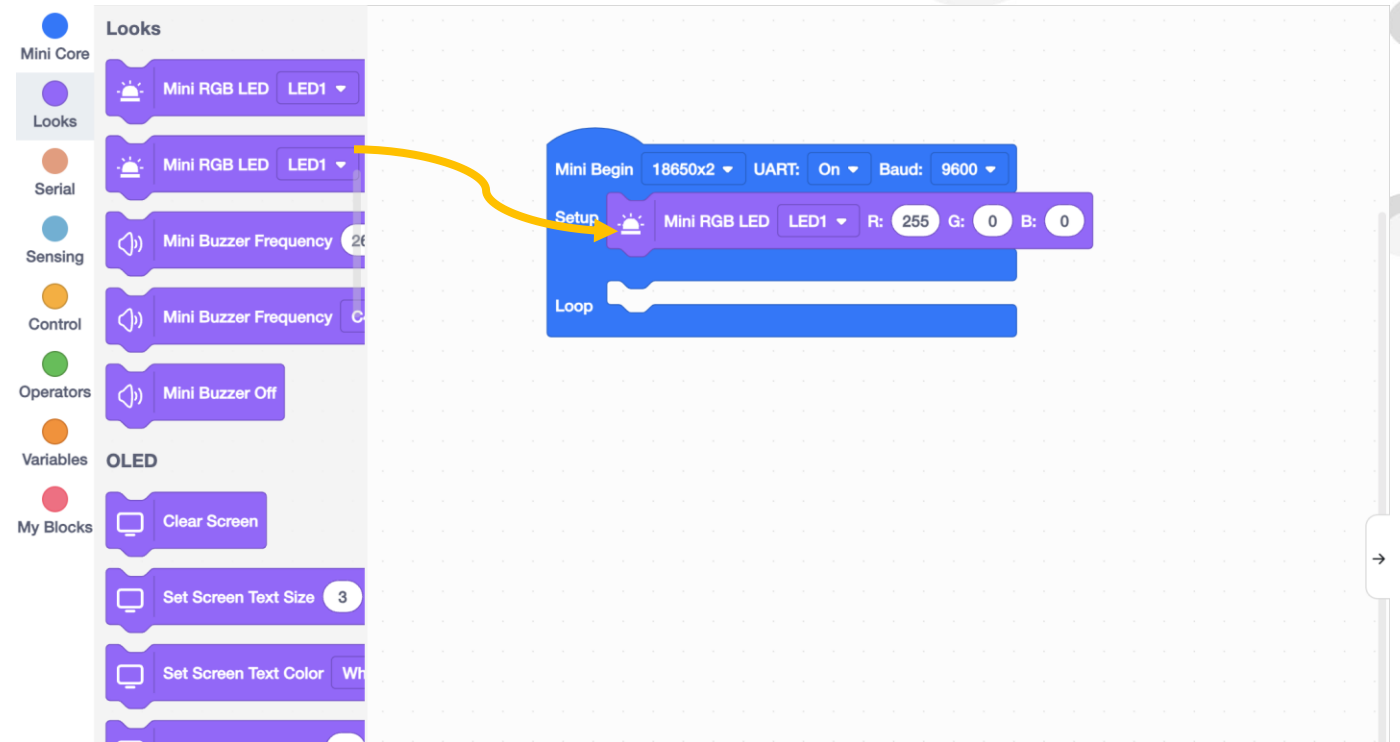
# Quick Start - Let RGBLED Lights Up

1. Drag and drop a **Mini Begin** block from the **Mini Core** category into the editing area.



# Quick Start - Let RGBLED Lights Up

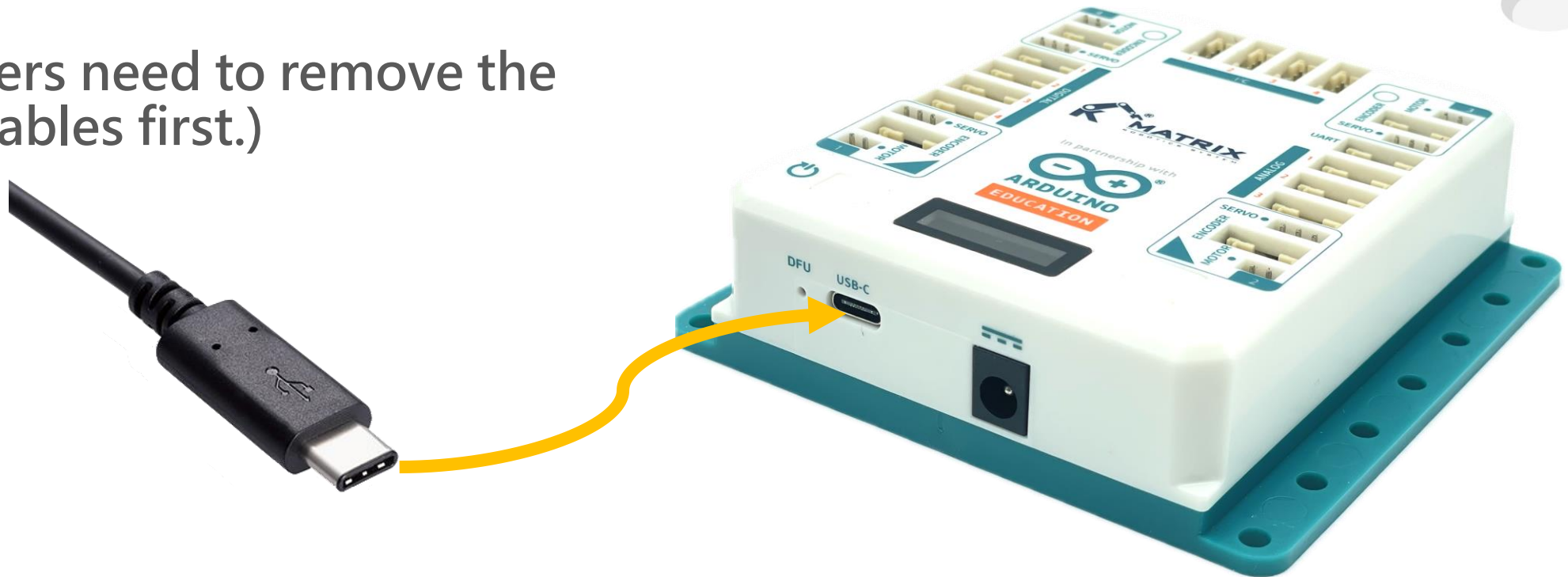
2. Drag and drop a **Mini RGB LED** block from the **Looks** category into the **Mini Begin**.



# Quick Start - Let RGBLED Lights Up

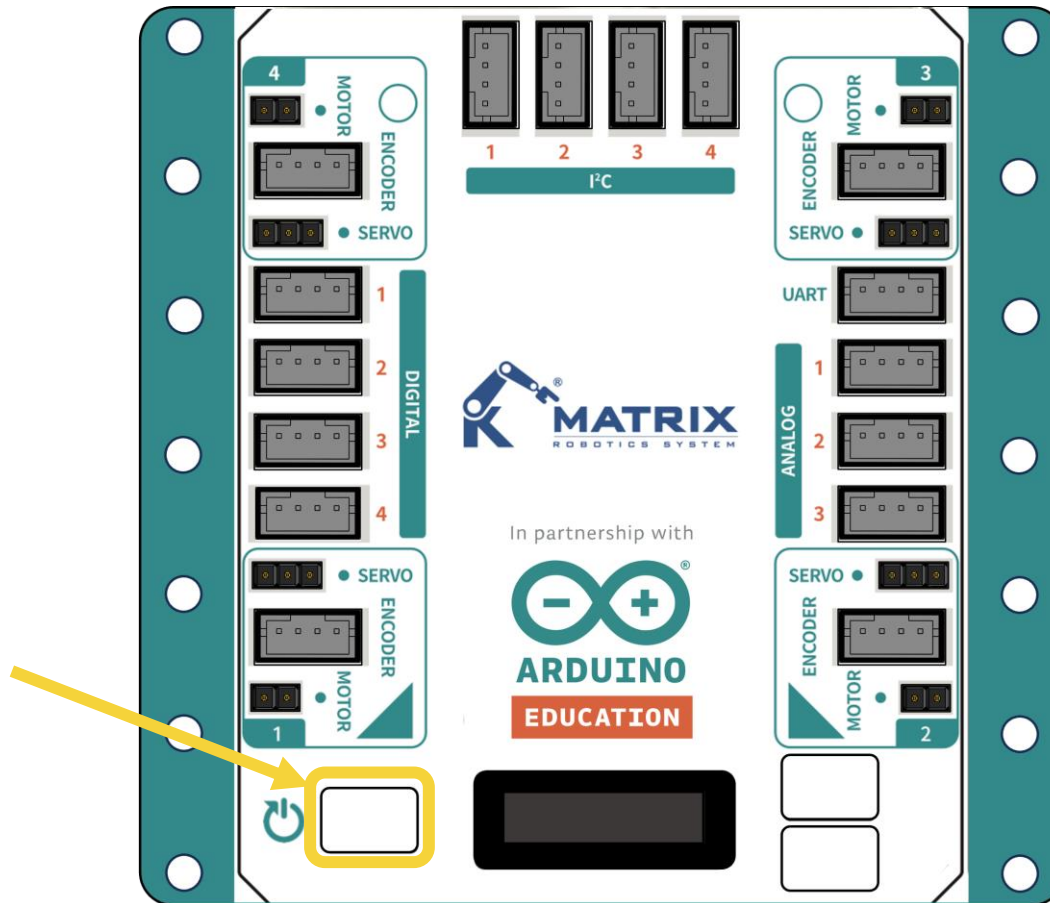
3. Connect the controller to the computer using a USB cable.

(Mac users need to remove the power cables first.)



# Quick Start - Let RGBLED Lights Up

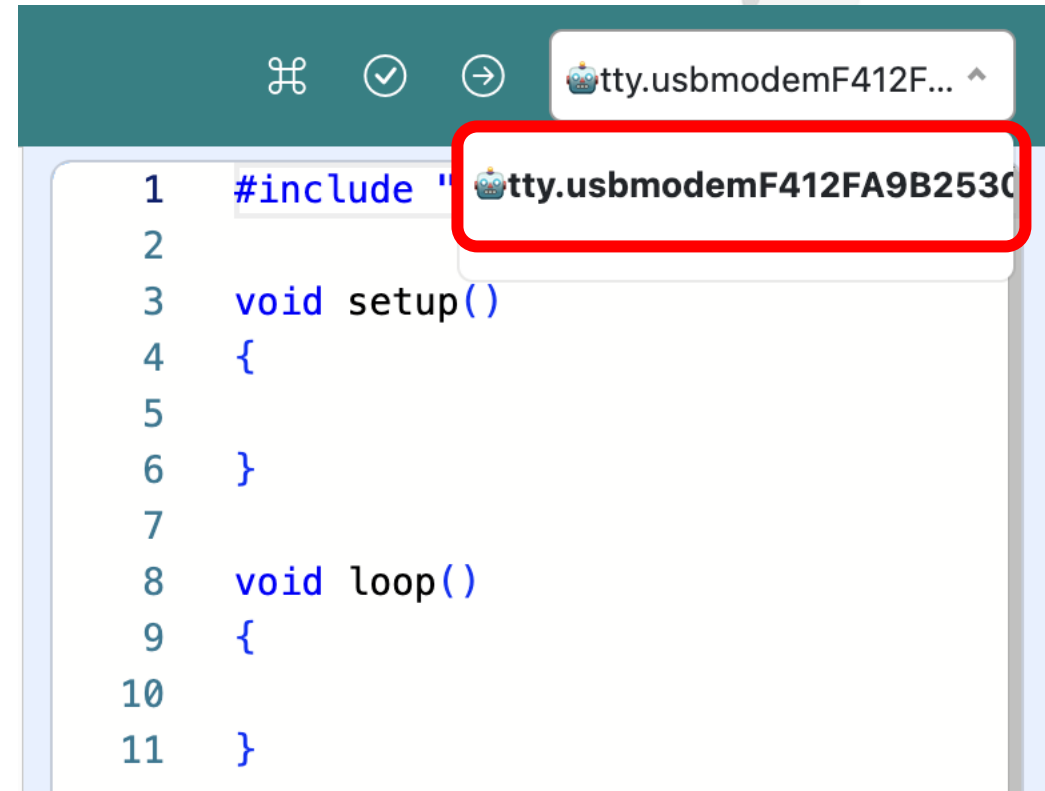
4. Press and hold the Reset Button to switch on the Mini R4.



# Quick Start - Let RGBLED Lights Up

5. Choose COM port ◦

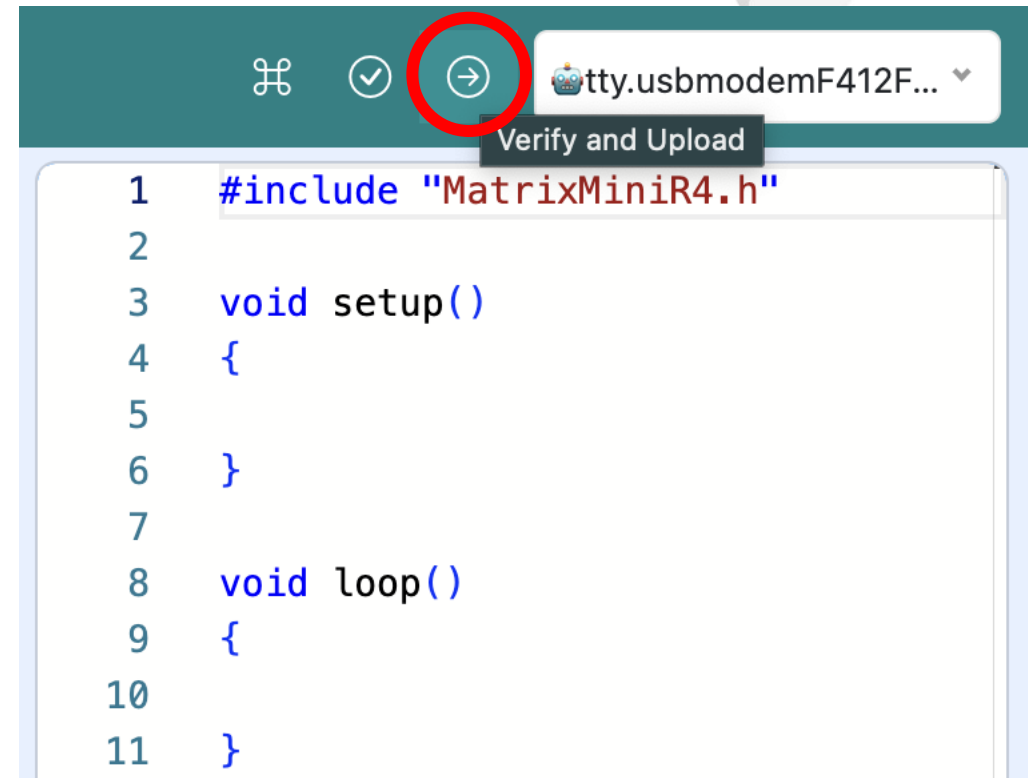
**\*\* The R4 controller is usually the one with the robot symbol in the COM list. \*\***



```
1  #include "tty.usbmodemF412FA9B253C"
2
3  void setup()
4  {
5
6  }
7
8  void loop()
9  {
10
11 }
```

# Quick Start - Let RGBLED Lights Up

6. Click "Verify and upload" in the upper right corner of the screen to upload the program to the MATRIX Mini R4 controller.



# Quick Start - Let RGBLED Lights Up

7. Wait for finish uploading.



**Compile/Upload Successful**

Done in 5.051 seconds  
reset()

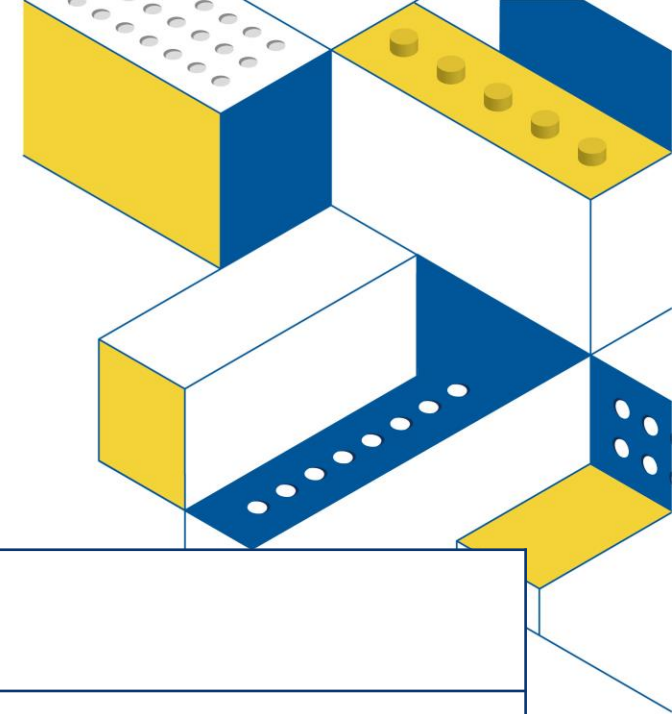
```
MatrixMiniR4 1.1.1 /Applications/MATRIXblock Mini
R4.app/Contents/arduino/libraries/MatrixMiniR4
SoftwareSerial 1.0 /Applications/MATRIXblock Mini
R4.app/Contents/arduino/packages/arduino/hardware/renesas_uno/1.2.1/libraries/SoftwareSerial
Wire /Applications/MATRIXblock Mini
R4.app/Contents/arduino/packages/arduino/hardware/renesas_uno/1.2.1/libraries/Wire
SPI /Applications/MATRIXblock Mini
R4.app/Contents/arduino/packages/arduino/hardware/renesas_uno/1.2.1/libraries/SPI
WiFiS3 0.0.0 /Applications/MATRIXblock Mini
R4.app/Contents/arduino/packages/arduino/hardware/renesas_uno/1.2.1/libraries/WiFiS3
```

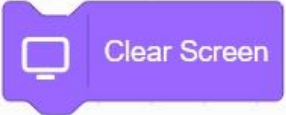
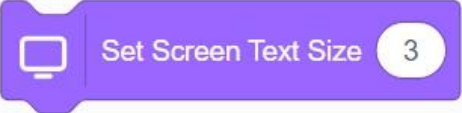
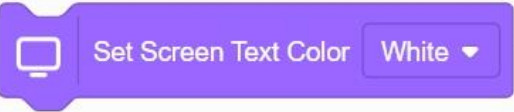
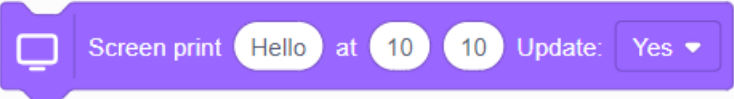
```
arduino:renesas_uno 1.2.1 /Applications/MATRIXblock Mini
R4.app/Contents/arduino/packages/arduino/hardware/renesas_uno/1.2.1
```

OK



# OLED Block Instructions

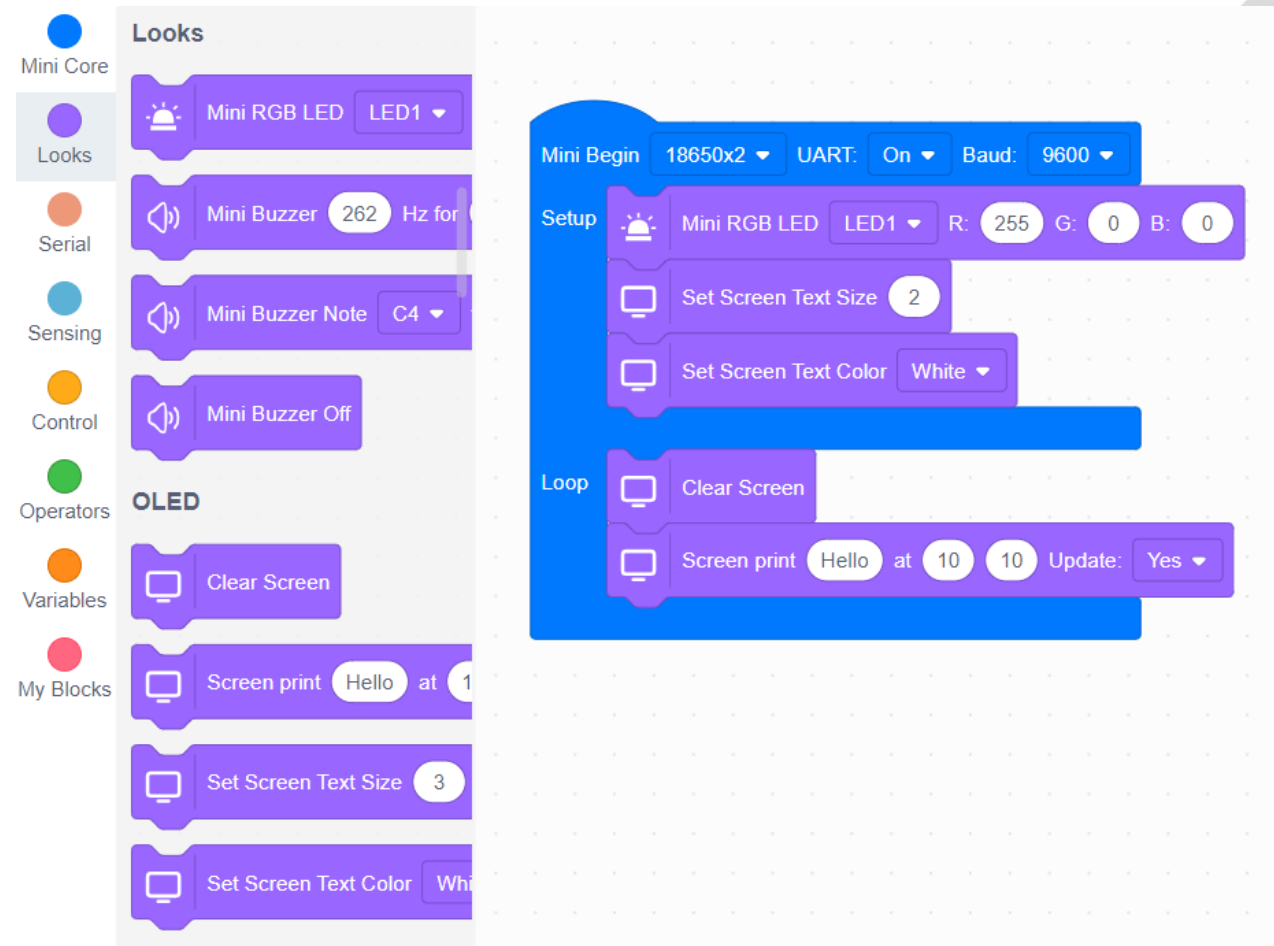


|                                                                                                                                                                                                                                                                                                                  |                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|  A purple block with a monitor icon and the text "Clear Screen".                                                                                                                                                                | Clear what is currently displayed on the screen.                                                                                              |
|  A purple block with a monitor icon, the text "Set Screen Text Size", and a circular input field containing the number "3".                                                                                                     | Set the size of the text to be displayed, the smallest is 1, usually placed in the setup. <b>(Default is 1)</b>                               |
|  A purple block with a monitor icon, the text "Set Screen Text Color", and a dropdown menu showing "White".                                                                                                                     | Setting the text color, usually placed in the setup. <b>(Default is White)</b>                                                                |
|  A purple block with a monitor icon, the text "Screen print", a text input field containing "Hello", the text "at", two circular input fields containing "10" and "10", the text "Update:", and a dropdown menu showing "Yes". | Print the text to the OLED, you can decide update screen or not for printing multiple content without “flashing”. The screen pixel is 128*32. |



# Quick Start -OLED Display

1. Drag a **Set Screen Text Size** block from the **Looks** category to the **Setup** and set the text size to 2.
2. Drag **Set Screen Text Color** blocks from the **Looks** category below the text size.
3. Drag and drop a **Clear Screen** and **Screen Print** block from the **Looks** category into the **Loop**.

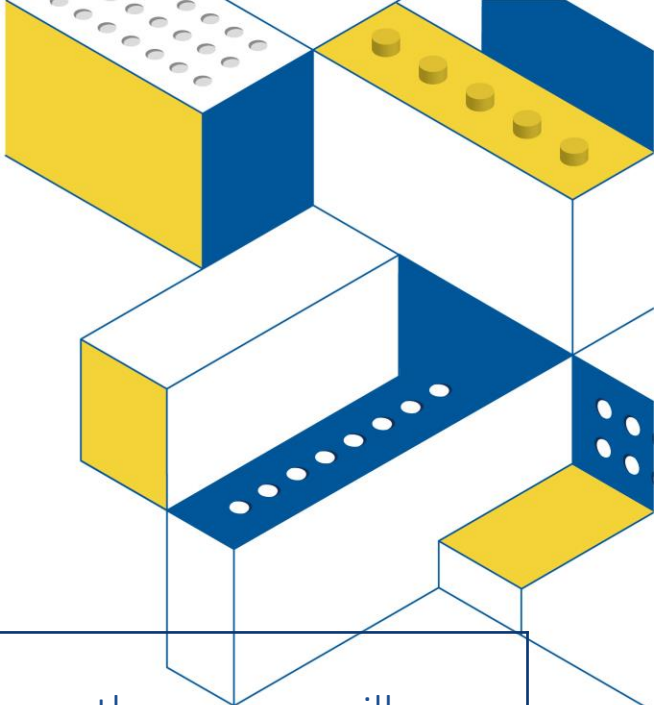


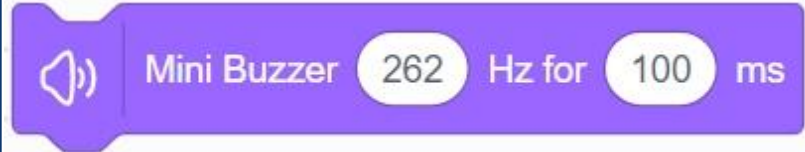
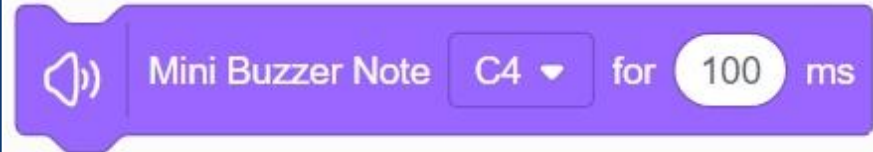
# Quick Start -OLED Display

4. Uploading the program and wait for finish uploading.



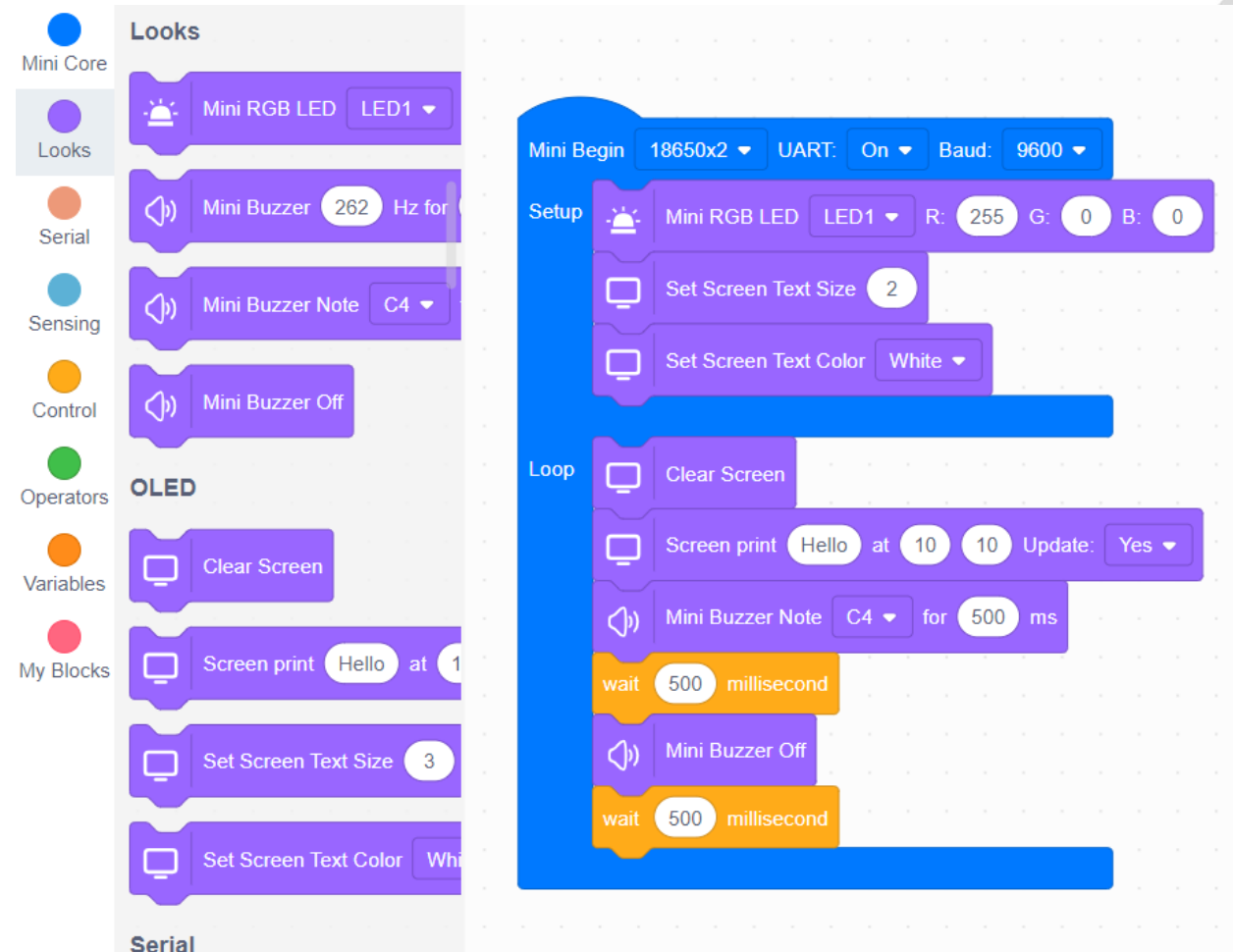
# Buzzer Block Instructions



|                                                                                                                                                                                                                                                                                                                           |                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|  A purple Scratch-style block with a speaker icon on the left. The text inside reads "Mini Buzzer", followed by a white circle containing the number "262", then "Hz for", another white circle containing "100", and finally "ms".      | Setting the frequency and playback time of Buzzer, the process will continue in parallel with the program below. |
|  A purple Scratch-style block with a speaker icon on the left. The text inside reads "Mini Buzzer Note", followed by a dropdown menu showing "C4" with a downward arrow, then "for", a white circle containing "100", and finally "ms". | Setting the chords and playback time of Buzzer, the process will continue in parallel with the program below.    |
|  A purple Scratch-style block with a speaker icon on the left. The text inside reads "Mini Buzzer Off".                                                                                                                                | Stop Playing Sound                                                                                               |

# Quick Start - Buzzer Play

1. Drag and drop a **Mini Buzzer Note** block from the **Looks** category into the loop of the Mini Begin and set the playback time to 500 milliseconds.
2. Wait for 500 milliseconds, then turn off the buzzer for 500 milliseconds.
3. The buzzer will be switched on and off in the loop.





**Congratulations on  
completing this unit!**

**Please click on the next unit!**